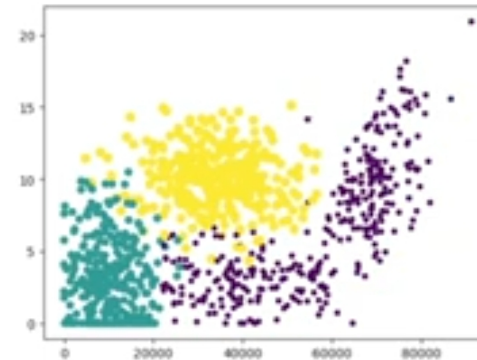


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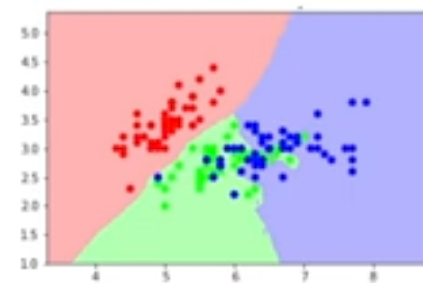
K-Nearest Neighbors (KNN) - intuition



K-Nearest Neighbors

About K-Nearest Neighbors:

1. Supervised Learning Model
2. Used for both Classification & Regression
3. Can be used for non-linear data
4. K - Neighbors



K-Nearest Neighbors

Classification Problem:



$K = 5$

● Didn't repay on time

● Repaid on time

● May not repay the loan on time

To Measure the distance between the data points:

❖ Euclidean Distance

❖ Manhattan Distance

K-Nearest Neighbors

Regression Problem:



$K = 5$

Salary of the person can be calculated as the mean of 5 nearest neighbors.

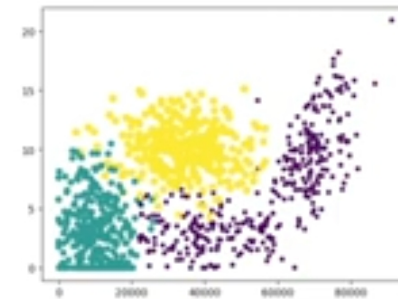
K-Nearest Neighbors

Advantages:

1. Works well with smaller datasets with less number of features
2. Can be used for both Classification & Regression
3. Easy to implement for Multi-class classification problems
4. Different distance criteria can be used
(eg: Euclidean Distance, Manhattan Distance)

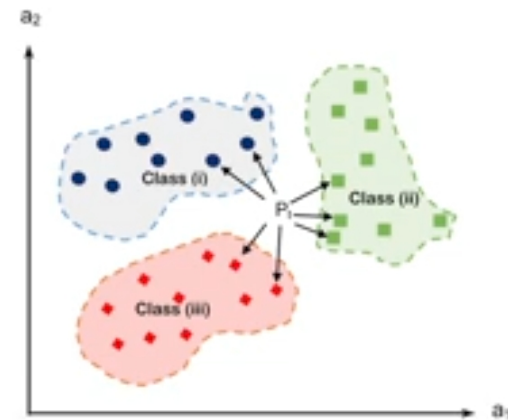
Disadvantages:

1. Choosing optimum "K" value
2. Less efficient with high dimensional data.
3. Doesn't perform well on imbalanced dataset

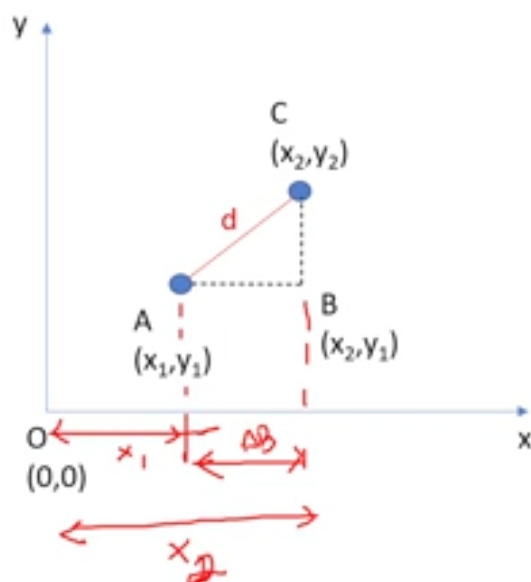


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Math behind K-Nearest Neighbors (KNN) Classifier



Euclidean Distance



Pythagoras Theorem:

$$AC^2 = AB^2 + BC^2$$

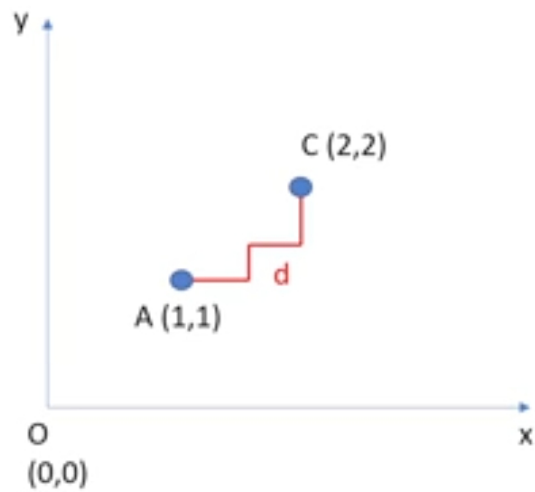
$$AC = \sqrt{AB^2 + BC^2}$$

$$AC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance "d" is called the Euclidean Distance.

Manhattan Distance



Manhattan Distance formula:

$$d = |x_1 - x_2| + |y_1 - y_2|$$

$$(x_1, y_1) = A (1,1)$$

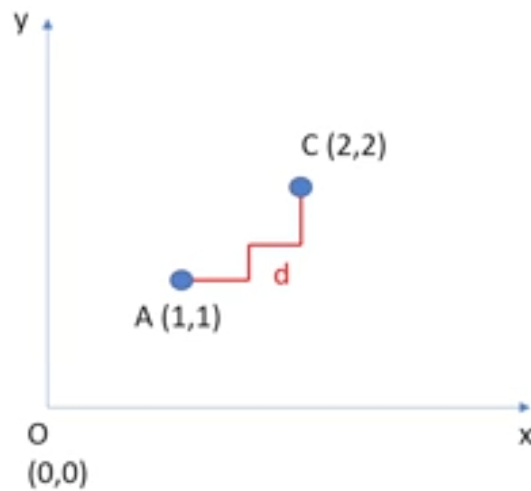
$$(x_2, y_2) = B (2,2)$$

$$d = |1 - 2| + |1 - 2|$$

$$d = 1 + 1$$

$$d = 2$$

Manhattan Distance



Manhattan Distance formula:

$$d = |x_1 - x_2| + |y_1 - y_2|$$

$$(x_1, y_1) = A (1,1)$$

$$(x_2, y_2) = C (2,2)$$

$$d = |1 - 2| + |1 - 2|$$

$$d = 1 + 1$$

$$d = 2$$

Manhattan distance is preferred over Euclidean distance when there is **high dimensionality** in the data.